Stat 145 (Multivariate Statistics)

Lesson 2.3. Testing Equality of Covariance Matrices

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Learning Outcomes

1. Derive the statistics for testing equality of covariance matrices.
2. Test hypotheses on equality of covariance matrices.

Introduction

One of the assumptions made when comparing two or more multivariate mean vectors is that the covariance matrices of the potentially different populations are the same. Before pooling the variation across samples to form a pooled covariance matrix when comparing mean vectors, it can be worthwhile to test the equality of the population covariance matrices. One commonly employed test for equal covariance matrices is Box's M-test.

Box's M Test

With t populations, the null hypothesis is

$$H_0 : \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \cdots = \boldsymbol{\Sigma}_t = \boldsymbol{\Sigma}$$

where $\boldsymbol{\Sigma}_i$ is the covariance matrix for population i, $i = 1, 2, \cdots, t$ and $\boldsymbol{\Sigma}$ the presumed common covariance matrix. The alternative hypothesis is that at least two of the covariance matrices are not equal.

Assuming multivariate normal populations, a likelihood ratio statistic for test H_0 given by

$$\Lambda = \prod_{i=1}^{t} \left(\frac{|\boldsymbol{S}_i|}{|\boldsymbol{S}_p|} \right)^{(n_i-1)/2}$$

Here n_i is the sample size for the i^{th} group, $\boldsymbol{S}_i$ is the i^{th} group sample covariance matrix and $\boldsymbol{S}_p$ is the pooled sample covariance matrix given by

$$\boldsymbol{S}_p = \frac{(n_1-1)\boldsymbol{S}_1 + (n_2-1)\boldsymbol{S}_2 + \cdots + (n_t-1)\boldsymbol{S}_t}{\sum_{i=1}^{t}(n_i-1)}$$

Box's test is based on his χ^2 approximation to the sampling distribution of $-2ln\Lambda$. Setting $M = -2ln\Lambda$ gives

$$M = \left[\sum_{i=1}^{t}(n_i-1)\right] ln|\mathbf{S}_p| - \sum_{i=1}^{t}\left[(n_i-1)ln|\mathbf{S}_i|\right]$$

If the null hypothesis is true, the individual sample covariance matrices are not expected to differ too much and, consequently, do not differ too much from the pooled covariance matrix. In this case, the ratio of the determinants in the formula of Λ will all be close to 1, Λ will be near 1 and Box's M statistic will be small. If the null hypothesis is false, the sample covariance matrices can differ more and the differences in their determinants will be more pronounced. In this case will Λ be small and M will be relatively large.

Let

$$u = \left[\sum_{i=1}^{t}\frac{1}{n_i-1} - \frac{1}{\sum_{i=1}^{t}(n_i-1)}\right]\left[\frac{2p^2+3p-1}{6(p+1)(t-1)}\right]$$

where p is the number of variables and t is the number of groups. Then

$$C = (1-u)M$$

has an approximate χ^2 distribution with degrees of freedom given by

$$\nu = \frac{1}{2}p(p+1)(t-1)$$

At significance level , reject H_0 if $C \geq \chi^2_{\alpha,\nu}$.

Remark:

Box's approximation works well if each $n_i > 20$ and if p and t do not exceed 5.

Example 2.4.1

Consider the scores of 2 men and 32 women in four psychological tests that was used in *Example 2.2.1*. Suppose we want to test

$$H_0 : \mathbf{\Sigma}_F = \mathbf{\Sigma}_M$$

versus

$$H_1 : \mathbf{\Sigma}_F = \mathbf{\Sigma}_M$$

The sample variance-covariance matrices of male and female participants are:

$$\boldsymbol{S}_M = \begin{bmatrix} 5.192540 & 4.545363 & 6.522177 & 5.250000 \\ 4.545363 & 13.184476 & 6.760081 & 6.266129 \\ 6.522177 & 6.760081 & 28.673387 & 14.467742 \\ 5.250000 & 6.266129 & 14.467742 & 16.645161 \end{bmatrix},$$

and

$$\boldsymbol{S}_F = \begin{bmatrix} 9.136089 & 7.549395 & 4.863911 & 4.151210 \\ 7.549395 & 18.603831 & 10.224798 & 5.445565 \\ 4.863911 & 10.224798 & 30.039315 & 13.493952 \\ 4.151210 & 5.445565 & 13.493952 & 27.995968 \end{bmatrix}.$$

and the pooled sample covariance matrix is

$$\begin{aligned} \boldsymbol{S}_p &= \frac{(n_1 - 1)\boldsymbol{S}_M + (n_2 - 1)\boldsymbol{S}_F}{n_1 + n_2 - 2} \\ &= \begin{bmatrix} 7.164315 & 6.047379 & 5.693044 & 4.700605 \\ 6.047379 & 15.894153 & 8.492440 & 5.855847 \\ 5.693044 & 8.492440 & 29.356351 & 13.980847 \\ 4.700605 & 5.855847 & 13.980847 & 22.320565 \end{bmatrix} \end{aligned}$$

We have $n_1 = n_2 = 32$, $|\mathbf{S}_1| = 7917.675$, $|\mathbf{S}_2| = 58958.07$, and $|\mathbf{S}_p| = 27325.23$. Thus,

$$\begin{aligned} M &= \Big[\sum_{i=1}^{t}(n_i - 1)\Big] ln|\mathbf{S}_p| - \sum_{i=1}^{t}\Big[(n_i - 1) ln|\mathbf{S}_i|\Big] \\ &= \Big[(32-1) + (32-1)\Big] \times ln(27325.23) - \Big[(32-1) \times ln(7917.675) + (32-1) \times ln(58958.07)\Big] \\ &= 14.5606, \end{aligned}$$

$$\begin{aligned} u &= \left[\sum_{i=1}^{t}\frac{1}{n_i - 1} - \frac{1}{\sum_{i=1}^{t}(n_i - 1)}\right]\left[\frac{2p^2 + 3p - 1}{6(p+1)(t-1)}\right] \\ &= \left[\left(\frac{1}{32-1} + \frac{1}{32-1}\right) - \frac{1}{(32-1) + (32-1)}\right]\left[\frac{2(4^2) + 3(4) - 1}{6(4+1)(2-1)}\right] \\ &= 0.06935484, \end{aligned}$$

and

$$\begin{aligned} C &= (1-u)M \\ &= (1-0.06935484)(14.5606) \\ &= 13.55075. \end{aligned}$$

Note that C has approximate chi-square distribution with $\nu = \frac{1}{2}p(p+1)(t-1) = \frac{1}{2}[4(4+1)(2-1) = 10$ degrees of freedom. At $\alpha = 0.05$, we reject H_0 if

$$C \geq \chi^2_{0.05,10} = 18.30704$$

Therefore, the data is not sufficient to reject the null hypothesis. This means that the covariance matrices of the test scores of males and females in the four tests are statistically equal.

Example 2.3.2

Load the **iris** data and test if the covariance matrices of *sepal.length*, *sepal.width*, *petal.length*, and *petal.width* are homogeneous among the three species .